

Jenkins-Setup

Jenkins-CI/CD Setup:

Prerequisites:

- 1.AWS Ec2-server
- 2.Java
- 3.Jenkins

Use t2.medium

1.Launch New EC2 server in the AWS Cloud

The screenshot shows the AWS Management Console interface for an EC2 instance named 'Jenkins_server'. The instance is in a 'Running' state. The details panel is expanded, showing the instance summary. The instance ID is i-03737b074bd205158. The public IPv4 address is 54.174.0.231. The private IPv4 address is 172.31.91.102. The public IPv4 DNS is ec2-54-174-0-231.compute-1.amazonaws.com. The instance is in the us-east-1d availability zone. The instance is a t2.micro type. The instance is running on the Amazon Linux 2 AMI.

Name	Instance ID	Instance st...	Instanc...	Status check	Alarm status	Availability Zone	Public IPv4 D
Jenkins_server	i-03737b074bd2...	Running	t2.micro	-	No alarms	us-east-1d	ec2-54-174-0-231.compute-1.amazonaws.com

Instance: i-03737b074bd205158 (Jenkins_server)

Details | Security | Networking | Storage | Status checks | Monitoring | Tags

Instance summary info

Instance ID i-03737b074bd205158 (Jenkins_server)	Public IPv4 address 54.174.0.231 open address	Private IPv4 addresses 172.31.91.102
IPv6 address -	Instance state Pending	Public IPv4 DNS ec2-54-174-0-231.compute-1.amazonaws.com open address

2.Once EC2 Created, Goto Security Group section:

The screenshot shows the AWS Management Console interface for the same EC2 instance 'Jenkins_server'. The 'Security' tab is selected in the details panel. The security groups section shows a single security group: sg-002062dd7fb3789e2 (launch-wizard-5).

Name	Instance ID	Instance st...	Instanc...	Status check	Alarm status	Availability Zone
Jenkins_server	i-03737b074bd2...	Running	t2.micro	2/2 checks p	No alarms	us-east-1d

Instance: i-03737b074bd205158 (Jenkins_server)

Details | **Security** | Networking | Storage | Status checks | Monitoring | Tags

Security details

IAM Role -	Owner ID 447953722679	Launch time Mon Dec 05 2022 10:06:07 GMT+0530 (India Standard Time)
Security groups sg-002062dd7fb3789e2 (launch-wizard-5)		

3.Select Security Group, Open in new Tab:

Instances (1/1) [Info](#)

Find instance by attribute or tag (case-sensitive)

Instance ID = i-03737b074bd205158 [Clear filters](#)

Name	Instance ID	Instance st...	Instanc...	Status check	Alarm status	Availability Zone
Jenkins_server	i-03737b074bd205158	Running	t2.micro	2/2 checks p	No alarms +	us-east-1d

Instance: i-03737b074bd205158

Details **Security**

Security details

IAM Role

Security groups

sg-002062dd7fb3789e2

Inbound rules

Open Link in New Tab

Open Link in New Window

Open Link in New Private Window

Bookmark Link...

Save Link As...

Save Link to Pocket

Copy Link

Search Google for "sg-002062dd7fb3..."

This Frame

Inspect Accessibility Properties

Inspect

Launch time

Mon Dec 05 2022 10:06:07 GMT+0530 (India Standard Time)

4. Then select Edit Inbound Rules:

sg-002062dd7fb3789e2 - launch-wizard-5 [Actions](#)

Details

Security group name	Security group ID	Description	VPC ID
launch-wizard-5	sg-002062dd7fb3789e2	launch-wizard-5 created 2022-12-05T04:35:29.788Z	vpc-07b639e336701f7c6
Owner	Inbound rules count	Outbound rules count	
447953722679	2 Permission entries	1 Permission entry	

Inbound rules **Outbound rules** **Tags**

You can now check network connectivity with Reachability Analyzer [Run Reachability Analyzer](#)

Inbound rules (2) [Manage tags](#) [Edit inbound rules](#)

Filter security group rules

4. Click on Add rule

under Port Type select All Traffic and source as Anywhere-IPv4

EC2 > Security Groups > sg-002062dd7fb3789e2 - launch-wizard-5 > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

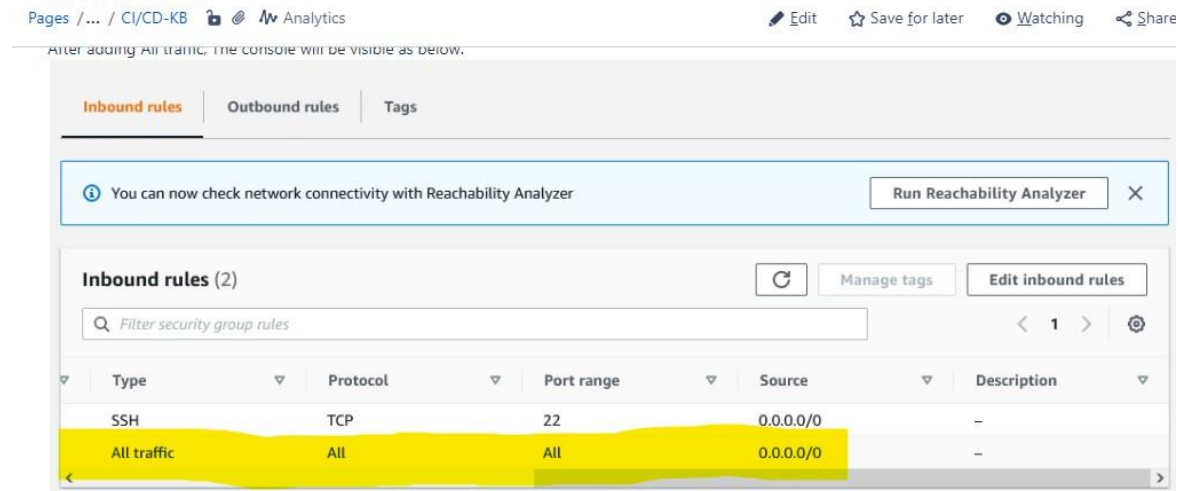
Security group rule ID	Type	Protocol	Port range	Custom	Description - optional
sgr-0fe5a37e61a100e51	SSH	TCP	22	Anywhere-IPv4	
sgr-0d57f118efb2d15fa	All traffic	All	All	Anywhere-IPv6	
				My IP	
				Anywh...	

[Add rule](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

and click on Save rules

After adding All traffic, The console will be visible as below.



Step 1: SSH Into Your Instance

First things first, let's SSH into the Amazon Linux instance where we're setting up Jenkins. Open up your terminal and run:

```
ssh -i <your_key>.pem ec2-  
user@<your_amazon_linux_instance_ip>
```

This command lets you log in to your instance.

Just

replace `<your_amazon_linux_instance_ip>` with your instance's IP address and `<your_key>` with the actual key name.

Example: **ssh -i jenkins-server-key.pem ec2-user@44.204.82.51**

All logged in? Perfect! Now, let's get down to business and start the Jenkins setup.

Step 2: Update the System

It's always a good practice to ensure that your system is up to date. Open a terminal window and enter the following command:

```
sudo dnf update
```

Step 3: Install Java

Jenkins is a Java application, so Java is a prerequisite. Install Java with the following command:

```
sudo dnf install java-17-amazon-corretto -y
```

Verify the installation with:

```
java -version
```

Step 4: Add the Jenkins Repository

Next, add the Jenkins repository to your system with the following commands:

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \  
https://pkg.jenkins.io/redhat-  
stable/jenkins.repo
```

Import a key file from Jenkins-CI to enable installation from the package:

```
sudo rpm --import https://pkg.jenkins.io/redhat-  
stable/jenkins.io-2023.key
```

Step 5: Install Jenkins

Now, install Jenkins by running:

```
sudo dnf install jenkins -y
```

Step 6: Start and Enable Jenkins

Enable and start the Jenkins service using:

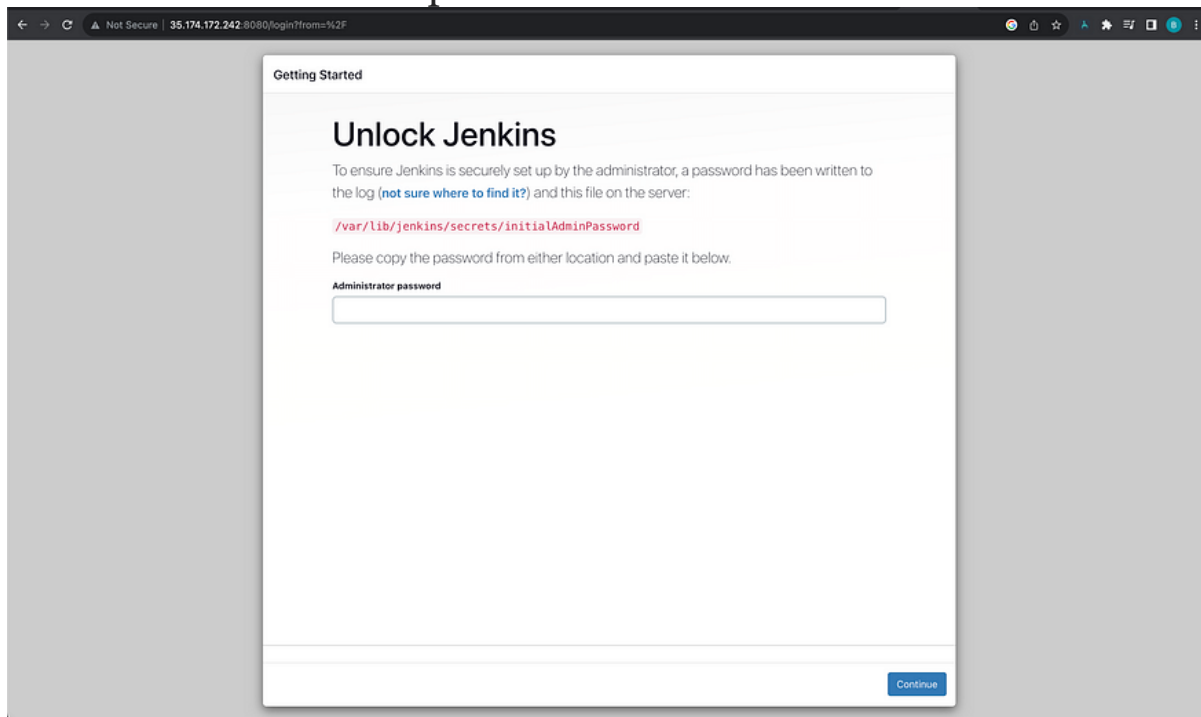
```
sudo systemctl enable jenkins  
sudo systemctl start jenkins
```

Step 7: Access Jenkins

Now, open your web browser and access Jenkins by navigating to:

```
http://your_amazon_linux_instance_ip:8080
```

You will see a setup wizard and be prompted to enter the administrator password.



##To get initial Admin password

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

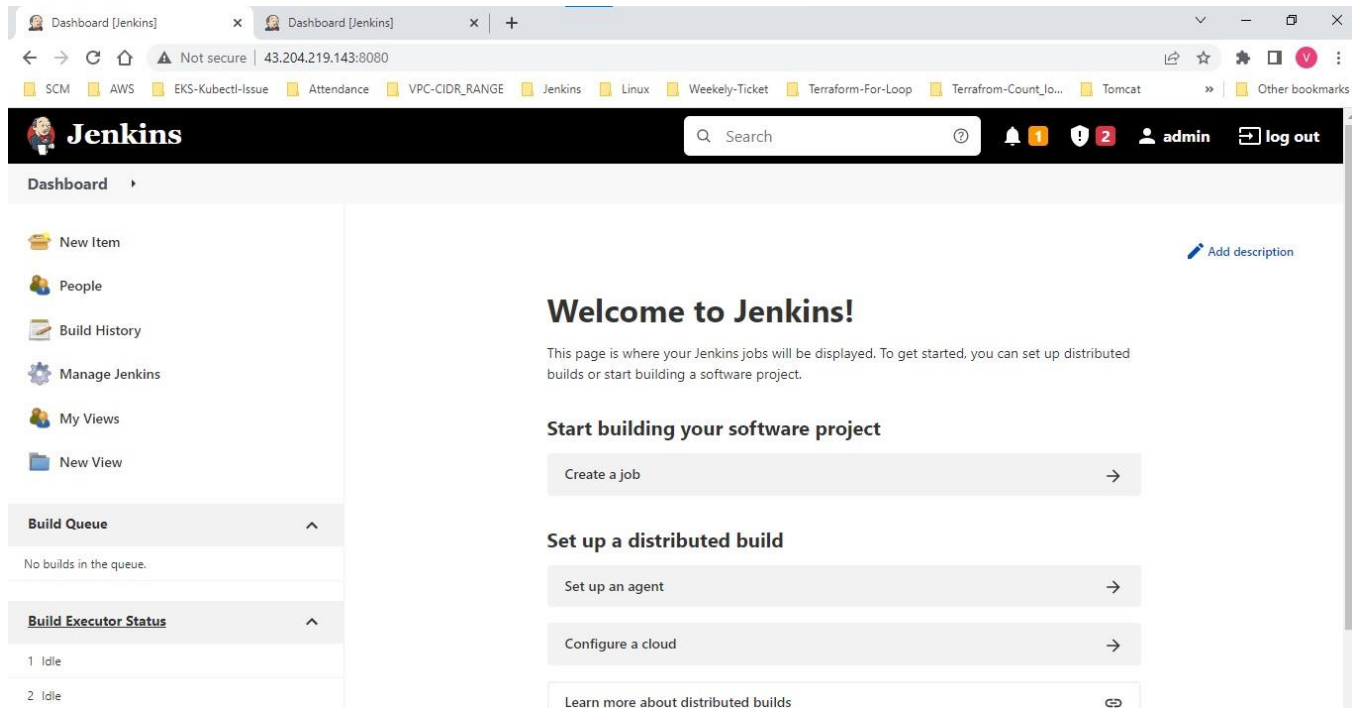


Signup with the username and passwd

username:admin

password:admin

email:admin@gmail.com



The screenshot shows the Jenkins web interface in a browser. The browser's address bar displays 'Not secure | 43.204.219.143:8080'. The Jenkins header includes the logo, a search bar, and user information for 'admin' with a 'log out' link. The left sidebar contains navigation links: 'New Item', 'People', 'Build History', 'Manage Jenkins', 'My Views', and 'New View'. Below these are sections for 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing two 'Idle' executors). The main content area features a 'Welcome to Jenkins!' message, a brief introduction, and three primary action buttons: 'Create a job', 'Set up an agent', and 'Configure a cloud'. A link to 'Learn more about distributed builds' is also present.

Dashboard [Jenkins] x Dashboard [Jenkins] x +

Not secure | 43.204.219.143:8080

SCM AWS EKS-Kubectf-Issue Attendance VPC-CIDR_RANGE Jenkins Linux Weekly-Ticket Terraform-For-Loop Terraform-Count_lo... Tomcat » Other bookmarks

Jenkins Search ? 1 2 admin log out

Dashboard ▸

- New Item
- People
- Build History
- Manage Jenkins
- My Views
- New View

Build Queue ^

No builds in the queue.

Build Executor Status ^

1	Idle
2	Idle

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job →

Set up a distributed build

Set up an agent →

Configure a cloud →

Learn more about distributed builds ↗